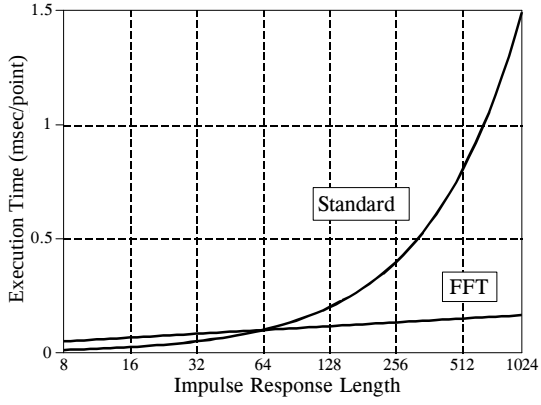

FIGURE 18-3
Execution times for FFT convolution. FFT convolution is faster than the standard method when the filter kernel is longer than about 60 points. These execution times are for a 100 MHz Pentium, using single precision floating point.

filter kernel. The crossover occurs when the filter kernel has about 40 to 80 samples (depending on the particular hardware used).

The important idea to remember: filter kernels shorter than about 60 points can be implemented faster with standard convolution, and the execution time is proportional to the kernel length. Longer filter kernels can be implemented faster with FFT convolution. With FFT convolution, the filter kernel can be made as long as you like, with very little penalty in execution time. For instance, a 16,000 point filter kernel only requires about *twice* as long to execute as one with only 64 points.

The *speed* of the convolution also dictates the *precision* of the calculation (just as described for the FFT in Chapter 12). This is because the round-off error in the output signal depends on the total number of calculations, which is directly proportional to the computation time. If the output signal is calculated *faster*, it will also be calculated more *precisely*. For instance, imagine convolving a signal with a 1000 point filter kernel, with single precision floating point. Using standard convolution, the typical round-off noise can be expected to be about 1 part in 20,000 (from the guidelines in Chapter 4). In comparison, FFT convolution can be expected to be an order of magnitude *faster*, and an order of magnitude more *precise* (i.e., 1 part in 200,000).

Keep FFT convolution tucked away for when you have a large amount of data to process and need an extremely long filter kernel. Think in terms of a *million* sample signal and a *thousand* point filter kernel. Anything less won't justify the extra programming effort. Don't want to write your own FFT convolution routine? Look in software libraries and packages for prewritten code. Start with this book's web site (see the copyright page).